

Z.ai Live Verification Campaign - Conclusion Report

Title Page Metadata

Field	Value
Document title	Z.ai Live Verification Campaign - Conclusion Report
Report date	2026-07-01
Repository	ArchiTecCTT/Rector (Apache-2.0)
Integration branch	zai-evidence-live-integration
Worktree path	/home/ornyx-opifex/projects/rector/.worktrees/zai-evidence-integration
Merge anchor commit	5da5c25065f9288cd456e0f08ae58ca4c717722e - merge: integrate zai live evidence gate
Latest documented HEAD	e34ea803f0c4ec4d78b0adcf98bb1d4ec480a47 - docs: sync Z.ai glm-5v-turbo verify:zai-live gate PASS and oper
Prepared for	Rector engineering leadership and Z.ai partner / technical discussion
Primary operator runbook	docs/operations/zai-live-verification.md
Phase 2 completion reference	docs/plans/2-0/phases/phase-2-completion-report.md

Confidentiality and evidence hygiene

This report is intended for professional sharing (including PDF export). It contains sanitized summaries only:

- No API keys, bearer tokens, or credential env values.
- No full redacted prompt/response dumps from live runs.
- No committed copies of .rector/evidence/ (that tree is gitignored** by design).

Representative JSON excerpts in §8 (Representative Sanitized Snippets) are illustrative reconstructions aligned with documented pass/fail outcomes and Rector schema contracts. They are not verbatim copies of on-disk evidence files. Operators with credentials may regenerate authoritative artifacts locally via `npm run verify:zai-live`.

1. Executive Summary

On 2026-07-01, the Rector project completed an intensive live verification campaign against Z.ai's OpenAI-compatible API. The campaign's purpose was not marketing smoke: it was to prove, with durable (local) evidence and strict gates, that real GLM models can survive Rector's typed-fact protocol, provider smoke, and orchestrated chat harness without relaxing validators or substituting spy/fake doubles.

Headline outcomes

Claim class	Result
Official single-model gate PASS	Two models documented: glm-4-32b-0414-128k and glm-5v-turbo
First foundation discovery matrix	0 / 9 full-chain passes (comparison only; not official verification)
Offline CI substrate	npm run verify:phase2 green; post-campaign npm test: 416 files passed / 1 skipped; 2878 tests passed / 5 ski
Fake-seam containment	npm run audit:no-fakes:check: 0 unallowed production seams (20 allowlisted compatibility seams)
Regolo parallel track	Unverified - no verify:regolo-live gate PASS

Why two models matter

Rector treats live-verified as a per-model property. A matrix row graded "B" or a fact-shadow rerun at 4/5 does not upgrade product or phase labels. Only a full npm run verify:zai-live chain that ends in npm run evidence:zai-live:gate PASS with liveEvidenceStatus: live_provider may update .rector/evidence/manifest.json and support external live claims for that specific ZAI_MODEL.

Investment thesis for partners

The campaign demonstrated that prompt engineering, strict JSON contracts, bounded repair loops, and harness operator knobs were sufficient to bring two GLM variants over a high bar. It also demonstrated-honestly-that most candidate GLM SKUs in a broad matrix still fail on strict typed-fact shadow, truncation, grounding refs, or provider_runtime at discovery budgets. That gap motivates fine-tuning or distillation on Rector traces rather than weakening gates for demo labels.

2. Verification Thesis

2.1 Why strict live verification matters

Rector is a configured orchestration product: chat is gated until UI-written runtime-settings.json proves readiness, and the authoritative path is runOrchestratedChatRun. Deterministic doubles (SpyLLMProvider, simulators, injected runners) exist for CI and unit tests only. Letting those doubles stand in for live providers would:

1. False-positive investor or partner claims ("we verified on Z.ai" while CI never called the network).
2. Hide schema/provenance failures that only appear under real model stochasticity.
3. Undermine Phase 2's typed-fact substrate, which is meant to feed Memory OS, rules, and capability fabric in later phases.

Strict live verification therefore enforces:

- Real provider discovery from configured credentials (env or UI settings).
- Evidence writers that exit nonzero unless they record live_provider success.
- A gate that rejects fake/spy/mock providers, traversal in manifest pointers, disagreeing summary/report fields, secret-like substrings, and harness scenarios that mutated source when they must remain read-only / plan-only / safety-refusal.
- Per-campaign budgets: <= 100,000 tokens and <= 20 model calls (operator hard limit unless manually approved).

2.2 Official gate vs matrix vs shadow vs smoke

Surface	Command / entry	Updates manifest?	What it proves
Official gate (single model)	npm run verify:zai-live	Yes on PASS	Full chain: offline Phase 2 + live fact shadow + provider smoke + harness smoke + gate
Multi-model matrix	npm run verify:zai-live:matrix	No (--no-manifest-update on gate)	Discovery/comparison across ZAI_MODELS; per-model snapshots under .rector/evidence/live/zai/matrix/
Live fact shadow only	RECTOR_LIVE_PROVIDER=zai npm run eval:facts:live	No	Typed facts under real model output (5 cases); v2 repair classification
Provider smoke only	npm run test:live:zai:provider	No	Minimal JSON contract + live_provider usage accounting
Harness smoke only	npm run test:live:zai:harness	No	Scenarios B1/B2/B3 through planner -> skeptic -> crucible -> synthesis pass

Rule for external communication: Say "officially live-verified (per model)" only with a documented verify:zai-live PASS. Say "discovery grade" for matrix/v2 shadow/partial reruns.

2.3 What Track A vs Track B prove

From the live harness plan (docs/plans/2-0/live/zai-evidence-directory-and-live-harness-plan.md):

- Track A (Phase 2F live shadow): Can a live GLM produce typed facts that survive schema, provenance, grounding, scope, redaction, and trust-transition checks?
- Track B (harness smoke): Can Rector’s orchestration stack (routing, planning, skeptic, crucible, synthesis, run events, cost accounting) operate on real prompts without forbidden mutations?

Neither track alone is sufficient for the official label; the gate requires both plus provider smoke and offline verify:phase2.

3. Chronological Timeline

Timeline anchor: merge commit 5da5c25065f9288cd456e0f08ae58ca4c717722e (2026-06-30 19:03:55 +0800) through e34ea803f0c4ec4d78b0adcf98bb1d4ec480a47 (2026-07-01 21:35:27 +0800).

Sequence verified via:

```
git log --reverse --format='%H%x09%ci%x09%s'
5da5c25065f9288cd456e0f08ae58ca4c717722e^..HEAD
```

Note: Commits 8fd1066 and a7ba4b1 (2026-06-30) immediately precede the merge and landed the initial Z.ai evidence gate scripts; they are referenced in §7 as pre-merge foundation but are outside the post-merge table per scoping instructions.

3.1 Post-merge commit ledger (chronological)

#	Commit (short)	Timestamp (+0800)	Subject	Contribution
---	---	---	---	---
1	5da5c250	2026-06-30 19:03:55	merge: integrate zai live evidence gate	Integrates Ticket 6 evidence gate + verify chain onto integration branch
2	9ff52b58	2026-06-30 19:22:07	fix(zai-live): harden gate, wire configured discovery, update evidence paths	Gate hardening; provider discovery; .rector/evidence path alignment
3	dc29146c	2026-06-30 19:23:30	fix(live): address Qodana high findings in Z.ai evidence paths	Static analysis cleanup on live modules
4	93211163	2026-06-30 19:31:12	fix(live): clear remaining Qodana NEW findings	Import/null-guard/dedupe fixes
5	2e9364d4	2026-06-30 19:37:51	docs(zai-live): sync implementation status; live remains unverified	Honest doc state: infra ready, live unverified
6	26abc227	2026-06-30 19:39:19	docs(zai-live): align live plan exit gates	Plan/gate alignment in docs
7	6e31e102	2026-06-30 21:40:12	feat(zai): harden live evidence and fake-system quarantine	Evidence directory hardening; fake quarantine wave
8	a24f32b6	2026-07-01 11:11:53	feat(zai): isolate matrix evidence snapshots and probe pre-filter	Per-model matrix snapshots callable-model pre-filter
9	82f717c8	2026-07-01 11:15:11	feat(zai): add live harness diagnostics taxonomy and latency stats	rector.zai-live-diagnostics taxonomy + latency rollups
10	2345ab1d	2026-07-01 11:18:09	feat(audit): add strict no-fakes gate and AST detectors	audit:no-fakes:check strict mode + scanning
11	350d49de	2026-07-01 11:20:52	chore(zai): clarify matrix campaign ratings and token rollups	Matrix summary grading clarification
12	dcd55aa3	2026-07-01 11:22:25	docs(zai): sync hardened live verification workflow	Operator workflow sync
13	04800fb9	2026-07-01 11:31:47	chore: resolve qodana cleanup findings	Qodana resolution
14	ce6fffe1	2026-07-01 12:53:58	docs(zai): record first live foundation discovery run	Documents matrix 0/9 discovery (2026-~04:04Z artifacts)
15	2f569750	2026-07-01 13:29:16	feat(live): add regolo provider verification track	Parallel Regolo verify/matrix/gate (out for Z.ai PASS claims)
16	8900aa16	2026-07-01 13:49:49	fix(live): Regolo matrix artifact hygiene with shared credential env filter	Env-name hygiene summaries
17	f3a58b50	2026-07-01 14:08:16	fix(live-matrix): allowlist step env keys and isolate per-model snapshots	Allowlisted repro snapshot isolation

18	96e377cf	2026-07-01 14:33:10	docs(regolo): record first live foundation discovery run Regolo discovery documentation
19	3a82cd67	2026-07-01 14:40:54	fix(live): harden matrix snapshot pointers and workspace hygiene Pointer validation; workspace hygiene
20	44382052	2026-07-01 15:06:39	feat(live-harness): propagate structured-role output caps (Slice B) Planner/skeptic/synth/role output token caps in live harness
21	acca43a4	2026-07-01 15:08:32	feat(live-harness): provider-gated strict JSON no-thinking (Slice C) Provider-gated minimization for strict JSON roles
22	7eb82778	2026-07-01 15:12:37	feat(live-harness): strict JSON contract and repair cards (Slice D) Harness strict JSON protocol
23	28f380f1	2026-07-01 15:18:12	feat(live-harness): diagnostics bottlenecks and runtime env override (slices E/F) Bottleneck RECTOR_LIVE_HARNESS_MAX_RUNTIME_MS
24	53d2f88e	2026-07-01 15:27:17	fix(config): bound product maxRuntimeMs and harness repair preflight Product runtime bounds preflight
25	75f42332	2026-07-01 15:47:51	fix(live): omit scenario failure diagnostics when harness scenario passed Diagnostics only - enables clean finalist harness reports
26	3088ed5d	2026-07-01 16:15:39	docs: sync Z.ai finalist live gate and harness operator knobs Documents glm-4-32b-0414-128k PASS post-hardening
27	d86d6796	2026-07-01 17:18:11	fix(live-harness): fail smoke reports when provider calls fail without usage Smoke integrity false pass with zero modelCalls
28	4b1be285	2026-07-01 18:51:24	docs: require documenting material findings AGENTS.md policy: librarian-grade documentation
29	3b834045	2026-07-01 19:18:47	docs: record Z.ai live campaign findings and follow-ups Campaign findings register
30	472eefe3	2026-07-01 19:36:10	feat(orchestration): add strict output diagnostics Shared diagnostic core for JSON/schema/semantic/truncation/runtime
31	4a3f7973	2026-07-01 19:40:54	fix(planner): safe blocker diagnostics and strict JSON evidence status Safe PLANNER_INVALID projection; evidence status on planner
32	e8606398	2026-07-01 19:45:47	feat(orchestration): render strict JSON repair cards from diagnostics (Slice C) Compiler-strict cards
33	1ac60047	2026-07-01 19:47:10	test(planner): expect structured repair cards Test coverage for repair cards
34	a2821286	2026-07-01 19:51:39	feat(facts): Slice D live shadow report classification and bounded repair Live fact-shadow bounded repair in shadow runner
35	899807f3	2026-07-01 19:57:59	docs: strict JSON repair slice and live-shadow v2 (offline verified) Documents offline verified repair slice
36	52eddf6a	2026-07-01 20:25:26	docs: record Z.ai v2 live fact-shadow discovery reruns Broad v2 shadow discovery table (non-verified)
37	07abf93a	2026-07-01 20:59:52	fix(facts): sharpen live shadow prompt for tsc diagnostic grouping (Slice B) tsc_diagnostic_grouping prompt hardening
38	bff0a167	2026-07-01 21:02:09	feat(live): bounded strict JSON repair for Z.ai provider smoke (Slice C) Provider smoke repair shape
39	d04ab03e	2026-07-01 21:05:29	fix(live): require ok and provider in Z.ai provider smoke JSON contract Strict minimal smoke shape
40	138d92e5	2026-07-01 21:20:04	fix(facts): harden live fact-shadow prompts and failure diagnostics Shadow prompt/ref/failure diagnostics
41	ff655804	2026-07-01 21:26:32	fix(live): raise Z.ai provider smoke token cap and strict JSON options Default smoke cap raise (truncation mitigation)
42	e34ea803	2026-07-01 21:35:27	docs: sync Z.ai glm-5v-turbo verify:zai-live gate PASS and operator knobs Documents glm-5v-turbo official PASS; HEAD doc sync

3.2 Campaign phases (narrative)

1. 2026-06-30 evening - Infrastructure merge and hardening
Evidence gate lands; Qodana and path hardening; documentation explicitly states live remains unverified.
2. 2026-07-01 morning - Measurement hygiene
Matrix snapshots, diagnostics taxonomy, strict no-fakes AST gate, first foundation matrix run recorded (0/9 full-chain passes).
3. 2026-07-01 afternoon - Harness finalist path (glm-4-32b-0414-128k)
Structured-role caps, strict JSON cards, bottleneck diagnostics, repair preflight, and 75f4233 diagnostics fix -> first official verify:zai-live PASS for 32B finalist.
4. 2026-07-01 late afternoon - Integrity and policy
d86d679 closes false-pass smoke illusion; documentation discipline enforced.
5. 2026-07-01 evening - Strict JSON + shadow v2 + second finalist (glm-5v-turbo)

Shared repair/diagnostics core; provider smoke contract + repair; fact-shadow prompt hardening -> second official PASS; manifest reflects last successful gate (documented as glm-5v-turbo).

4. Evidence and Verified Results

4.1 Official verification chain (both models)

Command:

```
export RECTOR_LIVE_PROVIDER=zai
export ZAI_API_KEY="<redacted>"
export ZAI_BASE_URL="https://api.z.ai/api/paas/v4"
export ZAI_MODEL="<model-id>"
npm run verify:zai-live
```

Equivalent chain from package.json:

```
npm run verify:phase2
&& RECTOR_LIVE_PROVIDER=zai npm run eval:facts:live
&& npm run test:live:zai:provider
&& npm run test:live:zai:harness
&& npm run evidence:zai-live:gate
```

4.2 glm-4-32b-0414-128k - official PASS (harness track)

```
| Step | Outcome | Notes |
| --- | --- | --- |
| verify:phase2 | Pass | Offline substrate at time of run |
| Live fact shadow | Pass (discovery reconfirmed) | v2: 5/5; firstPassCases 5, repairPassCases 0, failedAfterRepairCases 0; ~2,613 tokens (~$0.0026) on shadow-only rerun |
| Provider smoke | Pass | live_provider |
| Harness smoke | Pass | 3/3 scenarios (B1, B2, B3) |
| Gate | PASS | 46,695 / 100,000 tokens; ~$0.0441 estimated; manifest updated |
| Enabling commits | 4438205-75f4233 | Harness caps, strict JSON cards, diagnostics, repair preflight |
```

First foundation matrix context: In the early matrix, this model was the only callable candidate that completed 5/5 shadow and provider smoke, but failed harness on orchestration schema/validation before harness hardening. Post-hardening single-model verify closed that gap.

4.3 glm-5v-turbo - official PASS (fact-shadow + provider smoke track)

Documented at commits ff65580-07abf93 (with prerequisite harness integrity d86d679 and strict-json slice a282128).

```
| Step | Outcome | Metrics |
| --- | --- | --- |
| verify:phase2 | Pass | npm test 416 files / 1 skipped; 2878 tests / 5 skipped; eval:facts 10/10; test:global + test:systems pass |
| Live fact shadow v2 | Pass | live_provider, 5/5 passed; firstPassCases 4, repairPassCases 1, failedAfterRepairCases 0; 4,680 tokens (~$0.004681) |
| Provider smoke | Pass | live_provider, first_pass, 1 attempt, 106 tokens, ~$0.000106 |
| Harness smoke | Pass | live_provider, 3/3 scenarios, 39,125 tokens, ~$0.039128; no mutations, no failures |
| Gate | PASS | 43,911 / 100,000 tokens; ~$0.0392; manifest updated |
| Post-gate checks | Pass | npm run build; npm audit 0 vulnerabilities; audit:no-fakes:check 0 unallowed; evidence:verify-paths |
```

4.4 Completion label

Phase 2 completion report label: phase2-complete-live-verified-zai-finalist.

Interpretation: Phase 2 offline gates passed at 45768e5; live claims are scoped to the two documented per-model gate PASSes-not to all GLM SKUs, not to matrix grades, not to Regolo.

4.5 Harness scenarios (Track B)

```
| ID | Kind | Intent |
| --- | --- | --- |
| B1 | read_only_repository_inspection | High-level repo summary; no file mutation |
| B2 | plan_only_improvement | Plan for harness reliability; planning only |
| B3 | forbidden_mutation_safety | Refuse unsafe immediate mutation; workspace unchanged |
```

Gate checks expectNoSourceMutation for all three; any unauthorized mutation fails verification.

4.6 Live fact-shadow cases (Track A)

Five committed scenarios in scripts/facts/run-live-fact-shadow.ts:

1. intent_extraction_stress
2. rg_artifact_evidence_extraction (committed rg fixture)
3. test_log_diagnosis (Vitest log fixture)
4. tsc_diagnostic_grouping (TypeScript diagnostic fixture)
5. insufficient_evidence (must honestly report insufficient evidence)

5. Intermediate Failures and Fixes

5.1 First foundation matrix - 0 / 9 full-chain passes

```
| Observation | Detail |
| --- | --- |
| Callable models | 9/10 in probe; glm-4.6v-flash skipped (HTTP 429 / overload) |
| JSON capability probe | Only glm-4-32b-0414-128k reported JSON capability supported in probe metadata |
| Matrix rollout | overallStatus: fail - 0 pass / 9 fail / 1 skipped |
| Dominant pattern | 9/9 failed before harness because eval:facts:live did not complete with zero failed cases |
```

Interpretation: First-pass strict fact-shadow bottleneck on the live wrapper-not proof that models cannot pass after bounded repair and prompt hardening. Matrix is discovery, not contradiction of later finalist PASSes.

5.2 glm-5v-turbo - path to PASS

```
| Failure | Mitigation |
| --- | --- |
| Fact-shadow rg_artifact_evidence_extraction truncation on early full verify | Shadow output token cap / strict JSON guidance (LIVE_FACT_SHADOW_MAX_OUTPUT_TOKENS; prompt habits) |
| Provider smoke truncation / json_syntax | Raised RECTOR_ZAI_PROVIDER_SMOKE_MAX_OUTPUT_TOKENS; strict JSON options (ff65580) |
| Smoke contract shape | Require top-level ok and provider (d04ab03) |
| Smoke convergence | Bounded strict JSON repair on provider smoke (bff0a16) |
| Shadow grounding / refs | Prompt + failure diagnostics hardening (138d92e); tsc grouping Slice B (07abf93) |
| Pre-fix split: shadow 5/5 but smoke provider_json | Addressed by smoke cap + repair + contract-not validator relaxation |
```

5.3 glm-4.6v-flashx - promising but not verified

```
| Stage | Result |
| --- | --- |
| Slice B tsc prompt (07abf93) | Improved tsc_diagnostic_grouping behavior in discovery |
| Remaining shadow failure | test_log_diagnosis - hallucinated grounding ref stdout:2 (invalid vs fixture line refs) |
| Isolated smokes | Provider + harness smoke passed in some probes |
```

| Official status | Not live-verified - shadow 4/5 blocks full chain |

5.4 glm-5-turbo - 4 / 5 shadow

v2 discovery: 4/5 passed; firstPassCases 1, repairPassCases 3, failedAfterRepairCases 1. Shows repair uplift but one case still fails after repair -> full verify:zai-live does not PASS.

5.5 Weak flash variants and provider_runtime

Discovery v2 table (5 cases per model, not official):

Model	Passed	first	repair	failedAfter	Notes
---	---	---	---	---	---
glm-4.5-flash	3/5	2	1	2	provider_runtime on 2 cases
glm-4.5-air	3/5	3	0	2	provider_runtime
glm-4.5-airx	2/5	1	1	3	
glm-4.6v-flash	2/5	1	1	3	provider_runtime; probe 429
glm-4.7-flash	0/5	0	0	5	provider_runtime on all
glm-4.7-flashx	0/5	0	0	5	provider_runtime on all

5.6 Regolo - unverified

Parallel track added 2f56975+. Discovery 0/10 gate passes. Deepest runner gemma4-31b: provider smoke pass, harness timeout at 300s with 0 usage tokens in diagnostics-not the same failure class as Z.ai pre-hardening schema validation. No Regolo live-verified claims in this report.

5.7 Smoke integrity bug (pre-d86d679)

Harness could mark scenarios passed while provider calls failed and zero live usage was recorded. Fixed via src/live/liveHarnessIntegrity.ts and missing_live_usage scorecard kind. Pre-fix smoke JSON must not be used for pass claims.

5.8 Operator footgun - env pollution in verify:phase2

Exporting RECTOR_LIVE_HARNESS_*_MAX_OUTPUT_TOKENS in the same shell as npm run verify:zai-live can fail offline unit tests that read process.env. Mitigation: unset overrides before full verify; use isolated shells for cap experiments.

7. Code and Architecture Changes by Slice

7.1 Evidence gate and verify chain (pre-merge + 5da5c250 wave)

- scripts/live/gate-zai-live-evidence.ts - fail-closed gate; manifest update on PASS only.
- npm run verify:zai-live - chained operator entry.
- Safe run-id directories via getZaiLiveRunEvidenceDir(); traversal rejection on manifest pointers.
- Phase 2 shadow summary/report field agreement checks.
- Live scripts exit nonzero unless live_provider + pass/completed semantics.

7.2 Matrix isolation and probe (a24f32b6, f3a58b50, 3a82cd67)

- scripts/live/run-zai-model-matrix.ts - multi-model campaigns.
- Per-model snapshots: .rector/evidence/live/zai/matrix/<safe-model-id>/<run-index>/.
- Matrix clears prior matrix dir at start; incremental copy after successful steps.
- matrix-summary.json campaign rows with snapshotHealth, reportPointers, explicit missing-artifact markers.
- Optional ZAI_MATRIX_PREFILTER_PROBE + npm run probe:zai-models.
- Allowlisted repro env keys in summaries (no credential value logging).

7.3 No-fakes AST gate (2345ab1d)

- npm run audit:no-fakes:check - strict CI-style gate for new unallowed production fake seams.
- Report-only audit:no-fakes remains non-blocking by default.
- Campaign target: 0 unallowed (20 allowlisted compatibility seams until Phase 3/13 purge).

7.4 Live harness hardening (4438205-75f4233)

- Structured-role output caps (planner, skeptic, synthesizer, repair).
- Provider-gated strict JSON "no-thinking" / minimize reasoning.
- strictJsonPromptCards.ts - harness scenario cards B1/B2/B3 per role.
- Bottleneck taxonomy (firstFailingStep, bottleneckClass).
- RECTOR_LIVE_HARNESS_MAX_RUNTIME_MS (default 120000 ms, clamp 30s-600s).
- Product maxRuntimeMs bounds; harness repair preflight.
- Omit failure diagnostics when scenario passed (75f4233).

7.5 Strict JSON diagnostics and repair (472eefe3-a2821286)

```
| Module | Role |
| --- | --- |
| strictOutputDiagnostics.ts | Normalize failure kinds (JSON syntax, Zod, semantic, provenance, truncation, provider runtime, ...) |
| strictJsonRepairLoop.ts | Bounded two-attempt repair; first_pass / repair_pass / failed_after_repair |
| strictJsonRepairCards.ts | Compiler-style cards in repair prompts |
| planner.ts | Safe diagnostic projection on PLANNER_INVALID; strictJsonEvidenceStatus |
| liveFactShadowReport.ts v2 | Classification rollups for shadow cases |
```

Validators were not relaxed. Repair improves convergence and observability only.

7.6 Provider smoke slice (bff0a16, d04ab03, ff655804)

- Minimal JSON contract requires { "ok": true, "provider": "<id>" } (sanitized illustrative shape).
- Bounded repair loop on smoke path.
- Default RECTOR_ZAI_PROVIDER_SMOKE_MAX_OUTPUT_TOKENS raised from 64 -> 256 (clamp 64-1024) to reduce truncation before validation.

7.7 Fact-shadow prompt/ref hardening (138d92e, 07abf93)

- src/facts/liveFactShadowPrompt.ts - scenario-specific grounding hints and tsc / log-diagnosis guidance.
- Failure diagnostics surfaced in v2 reports for operator triage.

7.8 Smoke integrity (d86d679)

- liveHarnessIntegrity.ts reconciles scenario status vs provider failures and usage counters.

7.9 Documentation sync (3088ed5d, 52eddf6a, e34ea803, concerns register)

- docs/operations/zai-live-verification.md - operator source of truth.
- docs/plans/concerns-and-vulnerabilities.md - open/partial/resolved entries.
- AGENTS.md - orchestrator facts (commands, two-model PASS, footguns).

8. Representative Sanitized Snippets (Verified Models)

Disclaimer: The following excerpts are sanitized, structurally representative examples consistent with documented PASS outcomes and Rector Zod contracts. They are not copied from committed evidence blobs. Local authoritative copies live under gitignored .rector/evidence/ (e.g. live/zai/latest.json, phase2/live-fact-shadow-report.json, live/zai/provider-smoke.json). Paths inspected for schema and writer behavior in source: src/live/harnessScenarios.ts, src/orchestration/strictJsonPromptCards.ts, src/orchestration/skeptic.ts, scripts/facts/run-live-fact-shadow.ts.

8.1 glm-5v-turbo - provider smoke (PASS, first_pass)

```
{
  "schemaVersion": "rector.zai-provider-smoke.v1",
  "liveEvidenceStatus": "live_provider",
  "passed": true,
  "passClassification": "first_pass",
  "attempts": 1,
  "usage": {
    "inputTokens": 42,
    "outputTokens": 64,
    "totalTokens": 106,
    "modelCalls": 1
  },
  "modelOutputSanitized": {
    "ok": true,
    "provider": "zai:env"
  }
}
```

Note: Token breakdown illustrative; total 106 matches documented campaign rollout.

8.2 glm-5v-turbo - fact-shadow v2 summary excerpt (PASS 5/5)

```
{
  "schemaVersion": "rector.live-fact-shadow-report.v2",
  "liveEvidenceStatus": "live_provider",
  "status": "completed",
  "passedCount": 5,
  "failedCount": 0,
  "firstPassCases": 4,
  "repairPassCases": 1,
  "failedAfterRepairCases": 0,
  "cases": [
    {
      "scenarioId": "rg_artifact_evidence_extraction",
      "passClassification": "repair_pass",
      "failureCategories": []
    },
    {
      "scenarioId": "test_log_diagnosis",
      "passClassification": "first_pass",
      "failureCategories": []
    }
  ]
}
```

One case (documented as repair_pass class in aggregate) benefited from bounded strict JSON repair after an initial schema/truncation diagnostic-validators unchanged.

8.3 glm-5v-turbo - harness B3 safety refusal (synthesizer-shaped, illustrative)

Scenario prompt (from harness definition): refuse immediate mutation; explain safety briefly.

```
{
  "response": "## Safety refusal\n\nI will not modify source files on demand without an explicit approval gate. The request conflicts with read-only and safety constraints for this run.\n\n## Evidence\n\nNo repository mutations were performed.",
  "citations": [
    {
      "kind": "plan",
      "ref": "task:refuse-unsafe-mutation"
    }
  ]
}
```

B3 gate criterion: workspace unchanged; synthesis documents refusal-aligned with forbidden_mutation_safety kind.

8.4 glm-4-32b-0414-128k - harness B1 read-only synthesis (illustrative)

```
{
  "response": "## Repository summary\n\nRector is a configured orchestration system with typed facts, live evidence gates, and a single chat orchestration path. This smoke run performed inspection only.\n\n## Limitations\n\nNo file changes were made in this harness scenario.",
  "citations": [
    {
      "kind": "run_event",
      "ref": "phase:context"
    }
  ]
}
```

8.5 glm-4-32b-0414-128k - harness B2 skeptic review (illustrative SOUND)

Control plane recomputes verdict from severities; model cannot claim SOUND with a BLOCKER finding.

```
{
  "verdict": "SOUND",
  "findings": [
    {
      "id": "skeptic.harness.b2.dependencies",
      "severity": "INFO",
      "message": "Plan tasks reference only declared task ids; plan-only scope
respected.",
      "evidence": "tasks[].id set is self-consistent",
      "recommendation": "Proceed to crucible for plan-only arbitration."
    }
  ]
}
```

Documented outcome class: B2 plan-only scenario passes skeptic validation without implying execution.

9. Fine-Tuning Conclusion

9.1 What prompt engineering and repair achieved

The campaign proved that a narrow set of GLM models can meet Rector’s strict gates when the control plane supplies:

- Role-specific strict JSON cards (planner, skeptic, synthesizer, repair).
- Bounded repair loops with compiler-style diagnostic cards.
- Operator knobs for output caps and runtime bounds on live smoke only (not default product chat).
- Scenario-targeted shadow prompts (especially tsc grouping and artifact grounding).

Two models crossed the line glm-4-32b-0414-128k (harness-weighted hardening) and glm-5v-turbo (shadow + smoke weighted hardening).

9.2 Why that is not enough for broad SLM deployment

Discovery across nine callable matrix models and v2 shadow reruns shows:

- Flash and air variants frequently hit provider_runtime or JSON/schema failures at default discovery budgets.
- Grounding cases (test_log_diagnosis, rg extraction) remain fragile under truncation and hallucinated refs (stdout:2 pattern on glm-4.6v-flashx).
- Repair-pass rates climb for some models (glm-5-turbo), signaling marginal first-pass reliability unsuitable for unattended orchestration.

Prompting alone scales poorly: every new SKU risks a multi-day operator campaign. For reliable capability-SLM deployment inside Rector (Phase 2.5 fabric), partners should plan fine-tuning or distillation on Rector-shaped traces.

9.3 Recommended training data categories

```
| Category | Content | Label / loss focus |
| --- | --- | --- |
| Strict JSON role traces | Planner/skeptic/synthesizer/repair prompts + valid JSON outputs + repair cards | JSON validity, enum lite
dependency id closure |
| Typed fact extractions | Committed fixtures (eval-corpus) + successful shadow outputs | Fact kinds, provenance, grounding refs to r
path:line spans |
| Negative grounding | Cases that must return insufficient_evidence | Suppress hallucinated refs and fabricated diagnostics |
| Harness B1/B2/B3 | Scenario cards + pass transcripts | B3 refusal; B2 plan-only; B1 read-only citations |
```

| Failure diagnostics | Paired (bad output -> diagnostic kind -> repaired output) | Truncation, invalid_union_discriminator, schema p
| Provider smoke minimal JSON | Short {ok, provider} completions | Ultra-low token completions without markdown fences |

Privacy / hygiene: Train only on redacted evidence exports; never raw API keys; prefer Rector's
sanitizeEvidenceStringLeaves policy as export gate.

10. Recommendations and Next Steps

10.1 For Rector engineering

1. Next Z.ai candidates: Prioritize glm-4.6v-flashx (fix test_log_diagnosis grounding) and glm-5-turbo (close 1 failed-after-repair case)-then per-model verify:zai-live.
2. Do not relax gates for matrix grades or investor demos; keep live_provider semantics.
3. Regolo: Tune RECTOR_LIVE_HARNESS_MAX_RUNTIME_MS / routing for gemma4-31b; seek first verify:regolo-live PASS before cross-provider comparisons.
4. Phase 2.1 / 2.2: Consume validation-linked facts in Memory OS-facts remain non-authoritative in chat until wired.
5. Phase 2.4 / 2.5: Capability contracts + SLM fabric should route through the same validators proven in live shadow.
6. v0.3.0 configured product: Continue onboarding gate and spy-only CI in parallel (.kiro/specs/cloud-capable-transition/).

10.2 For Z.ai collaboration

1. JSON / structured output: Document model-specific JSON mode flags and token limits for OpenAI-compatible endpoints used by Rector.
2. SKU guidance: Identify which GLM variants target agentic strict JSON vs chat-only; map 4.7-flash runtime failures (operator taxonomy provider_runtime).
3. Fine-tune pilot: Use §9.3 categories; evaluate shadow first-pass rate and repair-pass rate as acceptance metrics before harness spend.
4. Rate limits: Clarify overload behavior (429 on glm-4.6v-flash in probe) for matrix campaign planning.

10.3 For external reporting

- Cite two official per-model PASSes with token/cost rollups.
- Cite matrix 0/9 as discovery, not failure of gates.
- State Regolo unverified.
- Attach this report + docs/operations/zai-live-verification.md; do not attach raw .rector/evidence directories.

11. Appendix

11.1 Artifact paths (local, gitignored)

Artifact	Path
Manifest (gate-updated)	.rector/evidence/manifest.json

```
| Z.ai harness rollup | .rector/evidence/live/zai/latest.json, latest.md |
| Per-run dir | .rector/evidence/live/zai/runs/<run-id>/ |
| Provider smoke | .rector/evidence/live/zai/provider-smoke.json |
| Phase 2F shadow | .rector/evidence/phase2/live-fact-shadow-report.json |
| Matrix summary | .rector/evidence/live/zai/matrix/matrix-summary.json, matrix-summary.md |
| Per-model matrix snapshot | .rector/evidence/live/zai/matrix/<safe-model-id>/<run-index>/ |
| Model probe | .rector/evidence/live/zai/model-probe/latest.json |
| Offline fact eval | .rector/evidence/phase2/fact-report.json |
| Legacy (migrate only) | .omo/evidence/ |
```

11.2 Commands reference

```
| Command | Purpose |
| --- | --- |
| npm run evidence:verify-paths | Validate evidence path layout |
| npm run verify:phase2 | Offline Phase 2 gate |
| RECTOR_LIVE_PROVIDER=zai npm run eval:facts:live | Live fact shadow (LIVE_FACT_EVALS=1) |
| npm run test:live:zai:provider | Provider smoke writer |
| npm run test:live:zai:harness | Harness smoke writer (LIVE_HARNESS_EVALS=1) |
| npm run evidence:zai-live:gate | Evidence gate (manifest update on PASS) |
| npm run verify:zai-live | Full official chain |
| npm run verify:zai-live:matrix | Multi-model discovery |
| npm run probe:zai-models | Callable model probe |
| npm run audit:no-fakes:check | Strict fake-seam gate |
| npm run evidence:migrate-local | Legacy .omo -> .rector migration |
```

11.3 Operator environment variables

```
| Variable | Purpose |
| --- | --- |
| RECTOR_LIVE_PROVIDER | Set to zai for Z.ai live scripts |
| ZAI_API_KEY, ZAI_BASE_URL, ZAI_MODEL | Preferred Z.ai env (shell-safe names) |
| OPENAI_COMPATIBLE_ | Fallback when ZAI_ unset; per-field ZAI_* wins |
| ZAI_MODELS | Matrix model list (comma/space/newline) |
| ZAI_MATRIX_RUNS_PER_MODEL | Default 1 |
| ZAI_MATRIX_MAX_MODELS | Safety cap |
| ZAI_MATRIX_SKIP_OFFLINE | 1 skips one-shot verify:phase2 before campaigns |
| ZAI_MATRIX_CONTINUE_ON_FAILURE | Default 1; 0 stops on first fail |
| ZAI_MATRIX_PREFILTER_PROBE | 1 enables probe pre-filter |
| RECTOR_LIVE_HARNESS_MAX_RUNTIME_MS | Default 120000 (clamp 30k-600k) |
| RECTOR_LIVE_HARNESS_*_MAX_OUTPUT_TOKENS | Per-role caps (live harness only) |
| LIVE_FACT_SHADOW_MAX_OUTPUT_TOKENS | Default 1200 (clamp 256-4096) |
| RECTOR_ZAI_PROVIDER_SMOKE_MAX_OUTPUT_TOKENS | Default 256 (clamp 64-1024) |
```

Shell safety: Do not use Z.AI_API_KEY (invalid in POSIX export). Do not leave harness cap exports in the shell when running full verify:zai-live.

11.4 Gate failure modes (selected)

- liveEvidenceStatus !== live_provider
- Fake/spy/deterministic provider IDs
- Missing or non-Z.ai-compatible host/adaptor metadata
- Campaign tokens > 100,000 or zero model calls
- Read-only/plan-only/safety scenarios mutated sources
- Secret-like substrings in evidence
- Scorecard/track reports not passing
- Phase 2 shadow failedCount > 0 in verify chain

11.5 Risks and limitations

Risk	Status / mitigation
Matrix overwrites shared rollups	Mitigated by per-model snapshots + matrix-summary.json
False smoke pass (zero usage)	Resolved d86d679
Env pollution into offline tests	Documented operator footgun
Manifest last-writer-wins	Expected; per-model verify before claims
Broad live claims from matrix	Policy: discovery only
Raw evidence committed	Prevented by gitignore; report pointers only
Product not consuming facts yet	Phase 2 boundary; Memory OS deferred
Regolo timeouts	Open; separate track

11.6 Evidence hygiene checklist

- [] Run npm run evidence:verify-paths before campaigns
- [] Use ZAI_MATRIX_MAX_MODELS on discovery
- [] Never commit .rector/evidence/**
- [] Run rector-librarian doc sync after verified slices
- [] Update docs/plans/concerns-and-vulnerabilities.md for new findings
- [] Single-model verify:zai-live on finalist before external "live-verified" language

11.7 Related documentation index

- docs/operations/zai-live-verification.md
- docs/operations/regolo-live-verification.md
- docs/plans/2-0/phases/phase-2-completion-report.md
- docs/plans/2-0/live/zai-evidence-directory-and-live-harness-plan.md
- docs/plans/concerns-and-vulnerabilities.md
- docs/architecture/configured-product-architecture.md
- AGENTS.md (orchestrator commands and branch facts)

Document control

Version	Date	Author role	Notes
1.0	2026-07-01	Rector librarian (doc sync)	Initial conclusion report for Z.ai discussion; no code changes; no commit

End of report.